

UCE303 :STRUCTURAL ANALYSIS

L	T	P	Cr
3	1	2	4.5

Course Objective: This course aims to develop an understanding of techniques of structural analysis to calculate displacements of statically determinate and indeterminate structures. This subject also introduces the concept of drawing influence line diagrams for determinate structures.

Displacements: Energy Methods: Strain energy in members, Betti's and Maxwell's Laws of reciprocal deflections, Castigliano's theorems, Unit load method for 2D-frames.

Indeterminate structures: Introduction, Static and kinematic indeterminacies, Stability of structures, internal forces in two and three-dimensional structures.

Analysis of Indeterminate Beams and Frames: Classical Methods: Theorem of three moments Methods of consistent deformation, Introduction to development of flexibility matrix; Conventional methods: Slope deflection method, Moment Distribution method, Introduction to development of stiffness matrix.

Moving Loads and Influence Line Diagrams for Statically Determinate Structures: Bending moment and shear force diagrams due to single and multiple concentrated rolling loads and uniformly distributed moving loads, equivalent UDL, shear force and bending moment envelopes.

Laboratory Work:

List of experiments:

1. To study the variation of BM at different locations in a simply supported beam.
2. To calculate deflections in a simply supported and cantilever beam.
3. To plot the deflected shape of a simply supported beam under constant load at centre as length varies.
4. To find the deflection of a pin connected truss.
5. Experimental project/assignment/micro-project: Students in a group of 4-5 students will do a mini-project based on modeling and analysis of indeterminate frames.

Course Learning Objectives (CLO)

Upon completion of this course, the students will be able to:

1. Calculate deformation of statically determinate structures.
2. Analyze statically indeterminate beams using classical and conventional methods.
3. Develop qualitative diagrams showing the displaced shape, bending moments and support reactions for an indeterminate structure.
4. Draw influence line diagrams for statically determinate beams.

Text Books

1. Structural Analysis, R. C. Hibbeler, Pearson Prentice Hall, Eighth edition, 2012.
2. Basic Structural Analysis, CS Reddy, Tata Mc Graw Hill, Third edition, 2011.

Reference Books

1. Advanced Structural Analysis, P. Dayaratnam, Tata McGraw Hill Publishers, 1983.
2. Theory of Structures, B.C. Punmia and A.K. Jain, Luxmi Publications 2003

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25
2	EST	35
3	Sessional: (May include the following) Assignment, Regular Lab assessment and Quizzes, Project (Including report, presentation etc.)	40